| Asset | No. | Threat Scenario | STRIDE | Attack Library No. |
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| E1.1. GPS/IMU | T1 | Manipulation of vehicle location perception via physical GNSS signal spoofing. | S | AL-32 |
| | T2 | Contamination of Position, Velocity, and Time (PVT) data via GNSS Spoofing. | S | AL-1 |
| | T3 | Induction of incorrect driving paths resulting from the bypass of GPS intrusion detection systems (IDS). | S | AL-31 |
| | T4 | Acceptance of malicious data due to the absence of authentication mechanisms for critical functions. | S | AL-12 |
| | T5 | Execution of physical control threats via portable spoofing devices demonstrated in verified attack cases. | S | AL-1 |
| | T6 | Disruption of sensor availability and integrity via jamming or spoofing utilizing commercial SDR tools. | S | AL-32 |
| | T7 | Ingestion of falsified GNSS measurements resulting from insufficient data authenticity verification. | S | AL-52 |
| | T8 | Injection of fraudulent location data facilitated by authentication bypass techniques such as message replay. | S | AL-19 |
| | T9 | Sensor data injection facilitated by inadequate endpoint constraints on communication channels. | S | AL-63 |
| | T10 | Failure in post-incident actor identification and non-repudiation due to the lack of logging for GNSS/IMU anomalies. | R | AL-89 |
| E1.2. Radar | T11 | Creation of ghost targets with fake range, velocity, and angle via FMCW radar spoofing. | S | AL-2 |
| | T12 | Generation of ghost objects in real-world environments via asynchronous spoofing without synchronization. | S | AL-66 |
| | T13 | Manipulation of object location perception via angle-based spoofing techniques. | S | AL-2 |
| | T14 | Execution of black-box physical layer attacks resulting from runtime estimation of frame and chirp parameters. | S | AL-2 |
| | T15 | Disruption of object detection (spoofing or concealment) demonstrated in ADAS radar systems. | S | AL-2 |
| | T16 | Acceptance of spoofed object data resulting from insufficient data authenticity verification. | S | AL-52 |
| | T17 | Injection of fake radar data facilitated by authentication bypass via spoofing techniques. | S | AL-19 |
| | T18 | Acceptance of unauthorized radar sources due to the absence of authentication mechanisms. | S | AL-12 |
| | T19 | Sensor data injection facilitated by inadequate endpoint constraints on communication channels. | S | AL-63 |
| | T20 | Failure in post-incident analysis and non-repudiation due to the lack of logging for radar anomalies. | R | AL-89 |
| E1.3. LiDAR | T21 | Creation of phantom obstacles via laser signal injection. | S | AL-33 |
| | T22 | Execution of phantom object attacks utilizing low-cost laser devices. | S | AL-33 |
| | T23 | Disruption of SLAM-based localization and mapping via asynchronous random point injection. | S | AL-34 |
| | T24 | Acceptance of spoofed point cloud data resulting from insufficient data authenticity verification. | S | AL-52 |
| | T25 | Injection of fake sensor data facilitated by authentication bypass techniques. | S | AL-19 |
| | T26 | Acceptance of unauthorized LiDAR sources due to the absence of authentication mechanisms. | S | AL-12 |
| | T27 | Sensor data injection facilitated by inadequate endpoint constraints on communication channels. | S | AL-63 |
| | T28 | Failure in post-incident actor identification and non-repudiation due to the lack of logging for LiDAR anomalies. | R | AL-89 |
| E1.4. UltraSonic | T29 | Creation of phantom obstacles with incorrect distance measurements via ultrasonic echo injection. | S | AL-35 |
| | T30 | Induction of random distance bias and continuous false positives via crosstalk or synchronization manipulation. | S | AL-35 |
| | T31 | Ingestion of falsified ultrasonic echo data resulting from insufficient data authenticity verification. | S | AL-52 |
| | T32 | Injection of fake sensor data facilitated by authentication bypass via spoofing techniques. | S | AL-19 |
| | T33 | Registration of malicious sensor devices due to the absence of authentication mechanisms for critical configuration functions. | S | AL-12 |
| | T34 | Acceptance of unauthorized ultrasonic sources due to inadequate authentication controls. | S | AL-12 |
| | T35 | Sensor data injection facilitated by inadequate endpoint constraints on communication channels. | S | AL-63 |

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| | T36 | Failure in post-incident actor identification and non-repudiation due to the lack of logging for ultrasonic anomalies. | R | AL-89 |
| E1.5. Camera | T37 | Induction of misclassification for signs/pedestrians via physical adversarial patches or stickers. | T | AL-36 |
| | T38 | Recognition of fake signs via digital phantom image injection (split-second projection). | S | AL-57 |
| | T39 | Detection failure (miss/false detection) of objects via physical adversarial examples. | T | AL-36 |
| | T40 | Perception failure via pixel distribution disruption using light/laser projection. | T | AL-37 |
| | T41 | Ingestion of manipulated video streams resulting from insufficient data authenticity verification. | S | AL-52 |
| | T42 | Acceptance of unauthorized video sources facilitated by authentication bypass via spoofing. | S | AL-19 |
| | T43 | Acceptance of external/fake camera feeds due to inadequate authentication mechanisms. | S | AL-12 |
| | T44 | Video frame injection/replay facilitated by inadequate endpoint constraints on communication channels. | S | AL-63 |
| | T45 | Failure in post-incident analysis and non-repudiation due to the lack of logging for frame anomalies. | R | AL-89 |
| E2. Vehicle Gateway & Actuators | T46 | Acceptance of forged control commands resulting from gateway/actuator identity spoofing. | S | AL-52 |
| | T47 | Injection of frames by external nodes facilitated by inadequate endpoint constraints on CAN/UDS/DoIP channels. | S | AL-52 |
| | T48 | Impersonation of legitimate nodes resulting from session/key theft due to vulnerable credential protection. | S | AL-14 |
| | T49 | Injection of unauthorized control commands via exploitation of remote gateway vulnerabilities. | S | AL-52 |
| | T50 | Tampering of control frames resulting from the absence or misconfiguration of message authentication (e.g., SecOC). | S | AL-19 |
| | T51 | Acceptance of manipulated ACK/status frames resulting from insufficient data authenticity verification. | S | AL-52 |
| | T52 | Impersonation of gateway roles or routing rules resulting from access control failures. | S | AL-12 |
| | T53 | Failure in actor tracing for abnormal control events due to the lack of logging. | R | AL-89 |
| E3. Driver | T54 | Bypass of voice authentication procedures via physical audio replay attacks. | S | AL-83 |
| | T55 | Unauthorized execution of commands via AI-based voice cloning or synthesis. | S | AL-54 |
| | T56 | Execution of unauthorized ADS mode transitions resulting from driver identity spoofing (e.g., Relay Attacks). | S | AL-81 |
| | T57 | Acceptance of fraudulent logins resulting from vulnerable session/key management. | S | AL-14 |
| | T58 | Bypass of driver approval procedures via exploitation of remote vulnerabilities in telematics or apps. | S | AL-65 |
| | T59 | Misinterpretation of external inputs as driver commands facilitated by inadequate HMI endpoint constraints. | S | AL-63 |
| | T60 | Acceptance of manipulated HMI events resulting from insufficient data authenticity verification. | S | AL-52 |
| | T61 | Assumption of driver roles by unauthorized users due to access control failures. | S | AL-12 |
| | T62 | Failure in non-repudiation for driver approval/handover due to the lack of logging. | R | AL-89 |
| | T63 | Acceptance of unauthorized sensor sources resulting from inadequate endpoint identification/authentication. | S | AL-12 |
| | T64 | Ingestion of falsified measurement data (NMEA/PointCloud) resulting from insufficient data authenticity verification. | S | AL-52 |
| | T65 | Propagation of physical sensor spoofing (Radar/LiDAR) into the driver layer. | S | AL-2, AL-33 |
| | T66 | Absence of entity/topic authentication due to missing or misconfigured DDS security plugins. | S | AL-63 |
| | T67 | Tampering of control/status frames resulting from the absence of message authentication (SecOC violations). | T | AL-19 |
| | T68 | Kernel memory corruption via out-of-bounds write or buffer copy vulnerabilities. | T | AL-62 |
| | T69 | Buffer overwrites via integer overflow or index validation errors. | T | AL-17 |
| | T70 | Kernel modification or DoS via Time-of-Check Time-of-Use (TOCTOU) race conditions. | T | AL-61 |
| | T71 | Kernel/driver modification via remote vulnerability chains (e.g., exploit chains). | T | AL-11 |

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| P1. Device Driver | T72 | Privilege escalation via insufficient permission management or configuration errors. | E | AL-29 |
| | T73 | Kernel privilege escalation via Use-After-Free or integer overflow exploitation. | E | AL-16 |
| | T74 | Code execution or privilege escalation via Jetson/L4T kernel/boot chain vulnerabilities. | E | AL-26 |
| | T75 | Unsigned code execution via physical paths such as RCM boot mode. | E | AL-22 |
| | T76 | Failure in tracking actor behavior due to insufficient logging of IOCTL/diagnostic events. | R | AL-89 |
| | T77 | Failure in auditing security events due to the absence of DDS security logging plugins. | R | AL-89 |
| | T78 | Exposure of metadata/keys via plaintext transmission or weak encryption. | I | AL-13 |
| | T79 | Information exposure via microarchitectural side-channels or uninitialized resources. | I | AL-64 |
| | T80 | Leakage of kernel addresses/keys via debug or management flaws. | I | AL-74 |
| | T81 | Driver/Kernel crash via NULL pointer dereference or memory leaks. | D | AL-78 |
| | T82 | Service denial (Lockup) via misconfiguration or synchronization flaws (e.g., uninitialized mutex). | D | AL-79 |
| | T83 | Denial of Service via memory corruption vulnerabilities like Use-After-Free. | D | AL-16 |
| | T84 | Driver layer saturation via CAN bus flooding attacks. | D | AL-51 |
| P2.1. OS | T85 | Creation of unauthorized processes, services, or nodes stemming from a lack of mutual authentication. | S | AL-63 |
| | T86 | Injection into data flows (DF2/6/7) facilitated by inadequate endpoint constraints (socket/IPC/topic binding errors). | S | AL-63 |
| | T87 | Ingestion of falsified frames resulting from insufficient data authenticity verification. | S | AL-52 |
| | T88 | Session hijacking and process impersonation caused by improper key management or cryptographic misconfiguration. | S | AL-14 |
| | T89 | Unauthorized access or control facilitated by the absence or misconfiguration of DDS Security plugins. | S | AL-63 |
| | T90 | Modification of Kernel/OS memory space via exploitation of Out-of-bounds or Integer Overflow vulnerabilities. | T | AL-62 |
| | T91 | Code execution or integrity violation via insufficient access control for IOCTL interfaces. | T | AL-80 |
| | T92 | Modification of configuration, log, or module files through Path Traversal vulnerabilities. | T | AL-65 |
| | T93 | Alteration of system configurations or execution of arbitrary commands via OS/Command Injection vectors. | T | AL-11 |
| | T94 | Corruption of kernel data and memory structures by exploiting vulnerabilities in the Linux CAN Broadcast Manager. | T | AL-62 |
| | T95 | Failure in tracking actor behavior due to insufficient logging of permission requests or I/O errors. | R | AL-89 |
| | T96 | Failure in auditing security events due to the absence of DDS logging plugins. | R | AL-89 |
| | T97 | Exposure of sensitive data resulting from transmission in plaintext or the use of weak cryptographic algorithms. | I | AL-13 |
| | T98 | Leakage of sensitive information via microarchitectural side-channels or uninitialized resources. | I | AL-64 |
| | T99 | Leakage of kernel addresses or debug information via kernel component flaws. | I | AL-74 |
| | T100 | Exposure of sensitive information via UEFI Management Mode side-channels. | I | AL-24 |
| | T101 | Kernel/OS service disruption via resource exhaustion or NULL pointer dereference. | D | AL-78 |
| | T102 | Saturation of network stack or driver paths via CAN flooding attacks. | D | AL-51 |
| | T103 | Limited Denial of Service via bootloader (MB2) errors. | D | AL-69 |
| | T104 | Denial of Service via SMMU or transaction handling flaws. | D | AL-70 |
| | T105 | Inadequate Privilege and Permission Management. (Includes insecure default settings, mismanagement of privileged accounts, and incorrect permission assignments.) | E | AL-29 |
| | T106 | Privilege escalation via exploitation of memory corruption (UAF/Integer Overflow). | E | AL-16 |
| | T107 | Arbitrary code execution via unsigned code download or unsafe deserialization. | E | AL-18 |

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| | T108 | Code execution or privilege escalation via boot chain (UEFI/RCM) vulnerabilities. | E | AL-22 |
| P2.2.1. Sensor Fusion | T109 | Acceptance of falsified frames (fake objects/location) resulting from insufficient data authenticity verification. | S | AL-52 |
| | T110 | Injection into Fusion topics/IPC facilitated by missing authentication for publishers/nodes. | S | AL-63 |
| | T111 | Misperception caused by the propagation of physical sensor spoofing/adversarial examples into the fusion layer. | S | AL-2, AL-33, AL-36 |
| | T112 | Bias in fusion results induced by GNSS spoofing (localization contamination) via data flow. | S | AL-1, AL-32 |
| | T113 | Manipulation of map/sensor parameters via path traversal vulnerabilities. | T | AL-65 |
| | T114 | Manipulation of fusion state/output via OS/Command injection. | T | AL-11 |
| | T115 | Code execution or object manipulation via unsafe deserialization of cache/map formats. | T | AL-18 |
| | T116 | Memory corruption or output distortion via numeric processing flaws (integer overflow/buffer copy). | T | AL-15, AL-17 |
| | T117 | State corruption or crash via Time-of-Check Time-of-Use (TOCTOU) race conditions. | T | AL-61 |
| | T118 | Failure in non-repudiation of attacks/malfunctions due to insufficient logging of sensor discrepancies. | R | AL-89 |
| | T119 | Failure in auditing authentication/permission events due to the absence of DDS security logging. | R | AL-89 |
| | T120 | Exposure of sensitive information (trajectory/history) in logs/cache via plaintext storage. | I | AL-20 |
| | T121 | Eavesdropping risk due to plaintext transmission or weak encryption in internal IPC/network. | I | AL-13 |
| | T122 | Leakage of memory/debug information via improper initialization or permission issues. | I | AL-27 |
| | T123 | Service stall or memory exhaustion via malicious inputs (e.g., random points, phantom objects). | D | AL-34 |
| | T124 | Overload of fusion processing paths via network/bus flooding attacks. | D | AL-51 |
| | T125 | Privilege escalation or arbitrary access via poor permission management. | E | AL-29 |
| | T126 | Arbitrary code execution via download/loading of unverified models/plugins/maps. | E | AL-48 |
| | T127 | Privilege escalation via exploitation of platform vulnerability chains (isolation bypass). | E | AL-26 |
| P2.2.2. Localization | T128 | Acceptance of spoofed PVT/Map data resulting from insufficient data authenticity verification. | S | AL-52 |
| | T129 | Induction of localization bias via GNSS spoofing (Position/Time contamination). | S | AL-1, AL-32 |
| | T130 | Acceptance of fake topics from unauthorized publishers/nodes due to missing authentication. | S | AL-63 |
| | T131 | Misleading pose estimation caused by the injection of random spoofing/ghost points into the matching/ICP process. | S | AL-34 |
| | T132 | Manipulation of a-priori data (HD Map/Parameters) via path traversal vulnerabilities. | T | AL-86 |
| | T133 | Manipulation of localization state/settings via OS/Command injection. | T | AL-11 |
| | T134 | Arbitrary code execution or state manipulation via unsafe deserialization of map/session snapshots. | T | AL-18 |
| | T135 | Memory corruption via integer overflow or buffer copy errors during numeric processing. | T | AL-17 |
| | T136 | State corruption or system crash via Time-of-Check Time-of-Use (TOCTOU) race conditions. | T | AL-61 |
| | T137 | Failure in non-repudiation of localization failures/spoofing due to insufficient logging. | R | AL-89 |
| | T138 | Failure in auditing authentication/permission events due to the absence of DDS security logging. | R | AL-89 |
| | T139 | Exposure of sensitive information (location, trajectory) via plaintext logging or file storage. | I | AL-13 |
| | T140 | Risk of eavesdropping on location/map metadata due to plaintext transmission in internal IPC/topics. | I | AL-13 |
| | T141 | Leakage of memory/register information via improper initialization or permission settings. | I | AL-64 |
| | T142 | Resource exhaustion (computation/memory) via malicious inputs (noise, random points, map replacement). | D | AL-34 |
| | T143 | Process crash (DoS) via unhandled exceptions or NULL pointer dereference. | D | AL-78 |

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| | T144 | Pipeline stall caused by repeated matching failures due to LiDAR random spoofing. | D | AL-34 |
| | T145 | Privilege escalation via insufficient permission management or access control. | E | AL-29 |
| | T146 | Arbitrary code execution via unverified download or deserialization of models/maps. | E | AL-18 |
| | T147 | Privilege escalation via exploitation of platform chain vulnerabilities (Boot/UEFI/RCM). | E | AL-22 |
| P2.2.3. InterPretation & Prediction | T148 | Injection of fake fusion results into InterPretation topics via unauthorized publisher/nodes. | S | AL-63 |
| | T149 | Acceptance of manipulated objects/tracks resulting from insufficient data authenticity verification. | S | AL-52 |
| | T150 | Impersonation of legitimate nodes via topic/IPC session hijacking caused by key/session management errors. | S | AL-63 |
| | T151 | Misclassification, detection failure, or tracking confusion via physical/digital adversarial examples. | T | AL-36, AL-38 |
| | T152 | Disruption of distance, velocity, and track consistency via LiDAR random spoofing or phantom points. | T | AL-34 |
| | T153 | Tampering of model parameters/weights or code execution via path traversal or unsafe deserialization. | T | AL-18, AL-40 |
| | T154 | Memory corruption or output manipulation via numeric processing flaws (integer overflow/buffer copy). | T | AL-17 |
| | T155 | State corruption or data damage via Time-of-Check Time-of-Use (TOCTOU) race conditions. | T | AL-61 |
| | T156 | Failure in non-repudiation of misjudgments due to insufficient logging of model versions/thresholds. | R | AL-89 |
| | T157 | Failure in auditing authentication/permission events due to the absence of DDS security logging. | R | AL-89 |
| | T158 | Exposure of sensitive information (object IDs, trajectories) via plaintext logs or dumps. | I | AL-13 |
| | T159 | Eavesdropping on Fusion/Tracking metadata via plaintext transmission in internal IPC/network. | I | AL-13 |
| | T160 | Leakage of memory addresses/registers via improper initialization or debug builds. | I | AL-64 |
| | T161 | Exposure of model keys/metadata via transient execution or UEFI side-channels. | I | AL-24, AL-64 |
| | T162 | Computation exhaustion or memory depletion via massive ghost objects or high-resolution frames. | D | AL-34, AL-102 |
| | T163 | Denial of Service via platform SMMU errors or driver faults. | D | AL-70 |
| | T164 | Privilege escalation via insufficient permission/access control management. | E | AL-29 |
| | T165 | Arbitrary code execution via unsafe deserialization or unverified downloads. | E | AL-18 |
| | T166 | Privilege escalation via exploitation of Jetson chain vulnerabilities (CBoot/RCM/NvGPU). | E | AL-22, AL-26 |
| P2.3. Planning | T167 | Acceptance of fake inputs resulting from missing or bypassed authentication for publishers/processes. | S | AL-63 |
| | T168 | Ingestion of manipulated inputs (Sensor/Status/Command) resulting from insufficient data authenticity verification. | S | AL-52 |
| | T169 | Interpretation of false Driver Command Codes masquerading as user intent via MLLM Prompt Injection. | S | AL-43 |
| | T170 | Manipulation of planning outcomes or arbitrary code execution via path traversal or unsafe deserialization. | T | AL-18, AL-65 |
| | T171 | Manipulation of planner state/output via OS/Command injection through environment variables. | T | AL-11 |
| | T172 | Distortion of decision-making (e.g., lane change/speed) via Adversarial ML or training data poisoning. | T | AL-40, AL-38 |
| | T173 | Compromise of planner integrity via exploitation of platform (Jetson) vulnerabilities. | T | AL-68 |
| | T174 | Failure in non-repudiation of path selection/constraints due to insufficient logging. | R | AL-89 |
| | T175 | Failure in auditing authentication/permission events due to the absence of DDS security logging. | R | AL-89 |
| | T176 | Exposure of sensitive information (future trajectory/commands) via plaintext logging. | I | AL-13 |
| | T177 | Eavesdropping on path/policy metadata via plaintext transmission or weak encryption. | I | AL-13 |
| | T178 | Leakage of memory/keys via platform side-channels or UEFI vulnerabilities. | I | AL-24, AL-64 |
| | T179 | Service stall or cycle violation caused by computational exhaustion from high-cost optimization (e.g., ghost objects). | D | AL-34 |

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| | T180 | Denial of Service affecting user processes via Jetson kernel/driver vulnerabilities. | D | AL-70, AL-78 |
| | T181 | Privilege escalation via insufficient permission management or excessive capabilities. | E | AL-29 |
| | T182 | Code execution or privilege escalation via exploitation of memory corruption (Buffer Overflow/UAF). | E | AL-16, AL-17 |
| | T183 | Isolation bypass and privilege escalation via Jetson NvGPU/Boot chain vulnerabilities. | E | AL-25, AL-69 |
| P2.4. Mode Manager | T184 | Acceptance of fake status/path/object info resulting from missing or bypassed authentication for publishers/processes. | S | AL-63 |
| | T185 | Ingestion of manipulated/replayed Vehicle Status or Planned Trajectory resulting from insufficient data authenticity verification. | S | AL-52 |
| | T186 | Masquerading as authorized nodes due to disabled or misconfigured DDS/ROS2 security (Authentication/Access Control). | S | AL-63 |
| | T187 | Injection of false alarms facilitated by session/key management errors in HMI authority/notification channels. | S | AL-14 |
| | T188 | Manipulation of mode policies via path traversal or unverified loading of integrity-unchecked files. | T | AL-65 |
| | T189 | Manipulation of mode decision/output via OS/Command injection through configuration or IPC. | T | AL-11 |
| | T190 | Internal object manipulation or code execution via unsafe deserialization of status caches/snapshots. | T | AL-18 |
| | T191 | Modification of status/output via memory corruption flaws (integer overflow/buffer copy). | T | AL-17 |
| | T192 | Failure in non-repudiation of mode transitions/handovers due to insufficient logging. | R | AL-89 |
| | T193 | Failure in auditing authentication events due to missing DDS security logging/tagging profiles. | R | AL-89 |
| | T194 | Difficulty in post-incident analysis due to failure to report errors/anomalies (e.g., auth failure, sensor mismatch). | R | AL-89 |
| | T195 | Exposure of sensitive information (authority, trajectory, status) via plaintext logging. | I | AL-13 |
| | T196 | Eavesdropping on authority/notification channels due to plaintext transmission or weak encryption. | I | AL-13 |
| | T197 | Service stall or cycle violation caused by resource exhaustion from massive false objects or status update flooding. | D | AL-34, AL-51 |
| | T198 | Blocking or delay of mode decisions via bus/network flooding (CAN/IPC/DDS). | D | AL-51 |
| | T199 | Unavailability of modules due to platform/driver DoS vulnerabilities. | D | AL-70, AL-78 |
| | T200 | Privilege escalation or unauthorized access via insufficient permission/ACL management. | E | AL-29 |
| | T201 | Code execution or privilege escalation via unsafe deserialization or unverified downloads. | E | AL-18 |
| | T202 | Isolation bypass and privilege escalation via Jetson platform chain vulnerabilities. | E | AL-22, AL-26 |
| P2.5. Control | T203 | Acceptance of fake authority/path inputs facilitated by missing or bypassed authentication for publishers. | S | AL-63 |
| | T204 | Injection into control data flows (DF6/10/11) via inadequate endpoint constraints (Topic/IPC binding). | S | AL-63 |
| | T205 | Calculation of wrong control commands due to acceptance of manipulated/replayed authority or path data. | S | AL-52 |
| | T206 | Tampering or replay of control commands (DF6) via lack of message authentication (SecOC violations). | S | AL-19 |
| | T207 | Node impersonation via missing or misconfigured DDS Security (Authentication/Access Control). | S | AL-63 |
| | T208 | Modification of control policies/gains via path traversal vulnerabilities. | T | AL-65 |
| | T209 | Manipulation of control pipeline state/output via OS/Command injection. | T | AL-11 |
| | T210 | Internal object manipulation or code execution via unsafe deserialization of state snapshots. | T | AL-18 |
| | T211 | Memory corruption or output manipulation via numeric processing flaws (integer overflow/buffer copy). | T | AL-17 |
| | T212 | Integrity loss or code execution via insufficient IOCTL access control in kernel/drivers. | T | AL-80 |
| | T213 | Failure in non-repudiation of control decisions due to insufficient logging of modes/thresholds. | R | AL-89 |
| | T214 | Failure in auditing authentication/permission events due to the absence of DDS security logging. | R | AL-89 |
| | T215 | Difficulty in post-incident analysis due to failure to report error states (e.g., auth failure, timeout). | R | AL-89 |

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| | T216 | Exposure of sensitive control metadata (gains, authority) via plaintext logging. | I | AL-13 |
| | T217 | Eavesdropping on control parameters via plaintext transmission or weak encryption. | I | AL-13 |
| | T218 | Leakage of sensitive addresses/keys via kernel/driver debug paths. | I | AL-74 |
| | T219 | Service stall or cycle violation via resource exhaustion from input flooding (e.g., false path updates). | D | AL-51 |
| | T220 | Delay or loss of control commands via CAN bus flooding/arbitration abuse. | D | AL-51 |
| | T221 | Unavailability of control module via platform/driver DoS vulnerabilities. | D | AL-70, AL-78 |
| | T222 | Privilege escalation or unauthorized access via insufficient permission/ACL management. | E | AL-29 |
| | T223 | Code execution or privilege escalation via exploitation of memory corruption (UAF/Integer Overflow). | E | AL-16, AL-17 |
| P2.6.1. Infotainment | T224 | Bypass of driver authentication via deepfake or replay voice attacks. | S | AL-54, AL-83 |
| | T225 | Injection into alert/audio channels facilitated by missing authentication for HMI/IPC. | S | AL-63 |
| | T226 | Misinterpretation of external inputs as driver/mode manager sources due to inadequate endpoint binding. | S | AL-63 |
| | T227 | Acceptance of manipulated HMI alerts or driver inputs resulting from insufficient data authenticity verification. | S | AL-52 |
| | T228 | Session hijacking or fake session creation resulting from vulnerable session/key management. | S | AL-14 |
| | T229 | Modulation of MLLM output (alerts/commands) via prompt injection or context contamination. | T | AL-3, AL-43 |
| | T230 | Modification of HMI assets (icons/text) via path traversal vulnerabilities. | T | AL-65 |
| | T231 | Manipulation of alert content or flow via OS/Command injection through inputs/environment variables. | T | AL-11 |
| | T232 | Code execution or internal object manipulation via unsafe deserialization of caches/plugins. | T | AL-18 |
| | T233 | Failure in tracking user/system behavior due to insufficient logging of voice/alert events. | R | AL-89 |
| | T234 | Failure in auditing authentication events due to the absence of DDS security logging. | R | AL-89 |
| | T235 | Difficulty in post-incident analysis due to failure to report errors (auth failure, ASR error). | R | AL-89 |
| | T236 | Exposure of sensitive information (voice, text, alerts) via plaintext logging. | I | AL-13 |
| | T237 | Eavesdropping on voice/text/status via plaintext transmission in internal channels. | I | AL-13 |
| | T238 | Leakage of memory/address information via improper initialization or debug builds. | I | AL-27 |
| | T239 | Service stall or cycle violation caused by audio/text flooding or large alerts. | D | AL-51 |
| | T240 | HMI rendering overload or freezing caused by alert flooding. | D | AL-51 |
| | T241 | Privilege escalation or unauthorized access via insufficient permission/ACL management. | E | AL-29 |
| | T242 | Arbitrary code execution via unverified plugin/resource download or unsafe deserialization. | E | AL-18 |
| | T243 | Isolation bypass and privilege escalation via Jetson Boot/UEFI/RCM vulnerabilities. | E | AL-22, AL-26 |
| | T244 | Masquerading as legitimate input via driver identity spoofing (deepfake/replay) or stolen sessions. | S | AL-54, AL-83 |
| | T245 | Session hijacking or fake session creation resulting from vulnerable session/key management. | S | AL-14 |
| | T246 | Injection into input flows (DF4.1/DF4.2) facilitated by inadequate endpoint constraints. | S | AL-63 |
| | T247 | Generation of malicious Driver Command Codes via Prompt Injection or Context Contamination. | T | AL-3, AL-43 |
| | T248 | Tampering of parameters or code execution via file path traversal, unverified loading, or unsafe deserialization. | T | AL-18, AL-65 |
| | T249 | Misjudgment of intent or induction of prohibited actions via adversarial inputs (Adversarial ML). | T | AL-38, AL-87 |
| | T250 | Manipulation of MLLM output/state via OS/Command Injection through environment variables or pipes. | T | AL-11, AL-45 |
| | T251 | Failure in non-repudiation due to insufficient logging of prompts, speakers, and safety filter decisions. | R | AL-89 |

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| P2.6.2. MLLM | T252 | Difficulty in post-incident analysis due to failure to report errors (e.g., ASR error, policy violation). | R | AL-89 |
| | T253 | Exposure of sensitive information (voice, intent, commands) via plaintext logging. | I | AL-13, AL-60 |
| | T254 | Eavesdropping on input/output metadata via plaintext transmission or weak encryption. | I | AL-13 |
| | T255 | Leakage of memory/key slots via improper initialization or debug builds. | I | AL-27, AL-64 |
| | T256 | Service denial (Stall) caused by resource exhaustion from token bombs or long audio inputs. | D | AL-8 |
| | T257 | Service denial via queue saturation caused by alert/input flooding. | D | AL-51 |
| | T258 | Privilege escalation or unauthorized access via insufficient permission/ACL management. | E | AL-29 |
| | T259 | Arbitrary code execution via unsafe deserialization or unverified model/plugin downloads. | E | AL-18, AL-48 |
| | T260 | Isolation bypass and privilege escalation via Jetson Boot/UEFI/RCM vulnerabilities. | E | AL-22, AL-26 |
| D1.1. Sensor Log & Evidence | T261 | Tampering of log evidence (overwrite/relocation) via path traversal or filename manipulation. | T | AL-65 |
| | T262 | Modification or deletion of logs by unauthorized entities due to inadequate access control/permissions. | T | AL-29 |
| | T263 | Inability to detect log tampering due to the absence of integrity protection (e.g., signatures, hash chains). | T | AL-13 |
| | T264 | Event record manipulation or code execution via unsafe deserialization in the log collection pipeline. | T | AL-18 |
| | T265 | Exposure of sensitive information (location, tokens) due to excessive logging in plaintext. | I | AL-13 |
| | T266 | Eavesdropping or leakage of logs due to plaintext storage/transmission (At-rest/In-transit). | I | AL-13 |
| | T267 | Leakage of memory/address information via improper initialization or debug configurations. | I | AL-27, AL-64 |
| | T268 | Service disruption or evidence loss caused by log flooding or disk exhaustion. | D | AL-51 |
| | T269 | Data loss (drop) due to I/O queue saturation from excessive event ingestion. | D | AL-51 |
| D1.2. Planning Log & Evidence | T270 | Tampering of planning logs (overwrite/relocation) via path traversal or filename manipulation. | T | AL-65 |
| | T271 | Modification or deletion of planning logs by unauthorized entities due to inadequate access control. | T | AL-29 |
| | T272 | Inability to detect log tampering due to the absence of integrity protection (signatures/hash chains). | T | AL-13 |
| | T273 | Event record manipulation or code execution via unsafe deserialization in the log collection pipeline. | T | AL-18 |
| | T274 | Exposure of sensitive information (future path, driver commands) due to excessive logging in plaintext. | I | AL-13 |
| | T275 | Eavesdropping or leakage of logs due to plaintext storage/transmission (At-rest/In-transit). | I | AL-13 |
| | T276 | Leakage of memory/address information via improper initialization or debug configurations. | I | AL-27, AL-64 |
| | T277 | Service disruption or evidence loss caused by log flooding or disk exhaustion. | D | AL-51 |
| | T278 | Data loss (drop) due to I/O queue saturation from excessive event ingestion. | D | AL-51 |
| D1.3. HMI Log & Evidence | T279 | Tampering of log evidence (overwrite/relocation) via path traversal or filename manipulation. | T | AL-65 |
| | T280 | Modification or deletion of HMI/MLLM logs by unauthorized entities due to inadequate access control. | T | AL-29 |
| | T281 | Inability to detect log tampering due to the absence of integrity protection (signatures/hash chains). | T | AL-13 |
| | T282 | Event record manipulation or code execution via unsafe deserialization in the log collection pipeline. | T | AL-18 |
| | T283 | Distortion of future HMI/LLM behavior via log poisoning (injection of malicious prompts/adversarial samples). | T | AL-40, AL-43 |
| | T284 | Exposure of sensitive information (voice, text, commands) due to excessive logging in plaintext. | I | AL-13 |
| | T285 | Eavesdropping or leakage of logs due to plaintext storage/transmission (At-rest/In-transit). | I | AL-13 |
| | T286 | Leakage of memory/address information via improper initialization or debug configurations. | I | AL-27, AL-64 |
| | T287 | Service disruption or evidence loss caused by log flooding or disk exhaustion. | D | AL-51 |

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| | T288 | Data loss (drop) due to I/O queue saturation from excessive event ingestion. | D | AL-51 |
| D1.4. Mode Manager Log & Evidence | T289 | Tampering of log evidence (overwrite/relocation) via path traversal or filename manipulation. | T | AL-65 |
| | T290 | Modification or deletion of mode logs by unauthorized entities due to inadequate access control. | T | AL-29 |
| | T291 | Inability to detect log tampering due to the absence of integrity protection (signatures/hash chains). | T | AL-13 |
| | T292 | Event record manipulation or code execution via unsafe deserialization in the log collection pipeline. | T | AL-18 |
| | T293 | Distortion of mode policies/alert rules via log poisoning (injection of malicious prompts/adversarial samples). | T | AL-40, AL-43 |
| | T294 | Exposure of sensitive information (authority, handover status) due to excessive logging in plaintext. | I | AL-13 |
| | T295 | Eavesdropping or leakage of logs due to plaintext storage/transmission (At-rest/In-transit). | I | AL-13 |
| | T296 | Leakage of memory/address information via improper initialization or debug configurations. | I | AL-27, AL-64 |
| | T297 | Service disruption or evidence loss caused by log flooding or disk exhaustion. | D | AL-51 |
| | T298 | Data loss (drop) due to I/O queue saturation from excessive event ingestion. | D | AL-51 |
| D1.5. Control Log & Evidence | T299 | Tampering of log evidence (overwrite/relocation) via path traversal or filename manipulation. | T | AL-65 |
| | T300 | Modification or deletion of control logs by unauthorized entities due to inadequate access control. | T | AL-29 |
| | T301 | Inability to detect log tampering due to the absence of integrity protection (signatures/hash chains). | T | AL-13 |
| | T302 | Event record manipulation or code execution via unsafe deserialization in the log collection pipeline. | T | AL-18 |
| | T303 | Exposure of sensitive control parameters (gains, limits, authority) due to excessive logging in plaintext. | I | AL-13 |
| | T304 | Eavesdropping or leakage of logs due to plaintext storage/transmission (At-rest/In-transit). | I | AL-13 |
| | T305 | Leakage of memory/address information via improper initialization or debug configurations. | I | AL-27, AL-64 |
| | T306 | Service disruption or evidence loss caused by log flooding or disk exhaustion. | D | AL-51 |
| | T307 | Data loss (drop) due to I/O queue saturation from excessive event ingestion. | D | AL-51 |
| DF1.1. Position, Motion Signal | T308 | Injection of NMEA/IMU frames via Man-in-the-Middle (MITM) or rogue nodes due to inadequate endpoint constraints. | S | AL-63 |
| | T309 | Acceptance of manipulated PVT/IMU samples resulting from insufficient data authenticity verification. | T | AL-52 |
| | T310 | Acceptance of replayed PVT/IMU frames due to the absence of freshness/anti-replay mechanisms. | S | AL-19 |
| | T311 | Tampering of frames in transit facilitated by the absence of message authentication (MAC/Signature). | T | AL-19 |
| | T312 | Eavesdropping on position/motion history due to plaintext transmission (UART/I²C/SPI). | I | AL-13 |
| | T313 | Interception and decryption of signals resulting from weak encryption or key management errors. | I | AL-13 |
| | T314 | Leakage of sensitive metadata (precision location/velocity) via bus tapping. | I | AL-13 |
| | T315 | Processing delay or data drop caused by frame flooding or IRQ saturation. | D | AL-51 |
| | T316 | Degradation of downstream services caused by GNSS jamming or error frame spikes. | D | AL-32 |
| DF1.2. Radar Signal | T317 | Injection or tampering of radar frames by third parties due to inadequate endpoint constraints on the link. | T | AL-63 |
| | T318 | Acceptance of manipulated Range-Doppler maps or object lists resulting from insufficient data authenticity verification. | T | AL-52 |
| | T319 | Acceptance of replayed radar frames due to the absence of freshness/anti-replay mechanisms. | T | AL-19 |
| | T320 | Processing of fake objects (Ghost Targets) as legitimate data caused by physical layer spoofing. | T | AL-2, AL-66 |
| | T321 | Tampering of Ethernet/CAN-FD frames facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T322 | Eavesdropping on object lists/sensor metadata due to plaintext transmission (LVDS/Ethernet/CAN). | I | AL-13 |
| | T323 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |

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| | T324 | Exposure of sensitive metadata (trajectory estimation, chirp parameters) via bus sniffing. | I | AL-13 |
| | T325 | Pipeline delay or data drop caused by frame flooding or IRQ saturation. | D | AL-51 |
| | T326 | Service degradation of the perception pipeline caused by interference, jamming, or crosstalk. | D | AL-32, AL-2 |
| DF1.3. Point Cloud Signal | T327 | Injection of Point Cloud frames via Man-in-the-Middle or rogue nodes due to inadequate endpoint constraints. | T | AL-63 |
| | T328 | Acceptance of manipulated XYZ(I) samples or headers resulting from insufficient data authenticity verification. | T | AL-52 |
| | T329 | Acceptance of replayed Point Cloud frames due to the absence of freshness/anti-replay mechanisms. | T | AL-19 |
| | T330 | Tampering of Ethernet/UDP frames facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T331 | Propagation of physical spoofing (ghost objects/random points) into the data flow. | S | AL-33 |
| | T332 | Eavesdropping on Point Cloud/Scan metadata due to plaintext transmission. | I | AL-13 |
| | T333 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T334 | Exposure of environmental data (map reconstruction potential) via sensitive metadata leakage. | I | AL-13 |
| | T335 | Pipeline delay or data drop caused by frame flooding or ring buffer saturation. | D | AL-51 |
| | T336 | Service degradation caused by sensor saturation or blinding (strong light/laser injection). | D | AL-33 |
| DF1.4. UltraSonic Range Signal | T337 | Injection or tampering of distance data frames via Man-in-the-Middle or rogue nodes due to inadequate endpoint constraints. | T | AL-63 |
| | T338 | Acceptance of fake echoes or biased distance data caused by physical spoofing or crosstalk. | S | AL-35 |
| | T339 | Acceptance of replayed distance frames due to the absence of freshness/anti-replay mechanisms. | S | AL-19 |
| | T340 | Tampering of digital link frames facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T341 | Eavesdropping on distance/status metadata due to plaintext transmission (UART/I²C/SPI). | I | AL-13 |
| | T342 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T343 | Exposure of sensitive patterns (parking habits/sensor state) via metadata leakage. | I | AL-13 |
| | T344 | Processing delay or data drop caused by frame flooding or IRQ saturation. | D | AL-51 |
| | T345 | Service degradation of the parking aid pipeline caused by ultrasonic interference or crosstalk. | D | AL-35 |
| DF1.5. Image Signal | T346 | Injection or tampering of frame data via Man-in-the-Middle due to inadequate endpoint constraints (Port/Bus binding). | T | AL-63 |
| | T347 | Acceptance of manipulated image frames (header/timestamp) resulting from insufficient data authenticity verification. | T | AL-52 |
| | T348 | Acceptance of replayed image frames resulting from the absence of freshness/anti-replay mechanisms. | T | AL-19 |
| | T349 | Tampering of digital link frames (CSI-2/GMSL) facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T350 | Propagation of fake objects (Phantom Frames) into the data flow via optical injection (Laser/LED). | S | AL-36, AL-57 |
| | T351 | Eavesdropping on video streams/metadata due to plaintext transmission (CSI-2/GMSL2/Ethernet). | I | AL-13 |
| | T352 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T353 | Exposure of PII (faces, license plates) and environmental data via video leakage. | I | AL-13 |
| | T354 | Pipeline delay or data drop caused by frame flooding or ring buffer saturation. | D | AL-51 |
| | T355 | Service degradation or pipeline failure caused by sensor blinding (strong light/laser). | D | AL-37 |
| DF2. Sensor | T356 | Injection or tampering of raw frames by unauthorized processes due to inadequate IPC/device node access control. | S | AL-63 |
| | T357 | Acceptance of manipulated or corrupted raw data frames resulting from insufficient data authenticity verification. | T | AL-52 |
| | T358 | Processing of replayed raw data buffers due to the absence of freshness/anti-replay mechanisms. | T | AL-19 |
| | T359 | Eavesdropping on raw data/metadata via plaintext transmission in internal channels (IPC/Debug ports). | I | AL-13 |

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| Raw Data | T360 | Interception and reuse of raw data resulting from weak encryption or key management errors. | I | AL-13 |
| | T361 | Exposure of sensitive information (precise location, object tracks) due to excessive inclusion in raw data. | I | AL-13 |
| | T362 | Service stall or data loss caused by frame/buffer flooding leading to IPC/queue saturation. | D | AL-51 |
| | T363 | Pipeline degradation caused by a storm of error frames or unhandled exceptions. | D | AL-51 |
| DF2.1. Environment Perception Data | T364 | Injection or tampering of perception data via unauthorized processes due to weak endpoint constraints (Socket/IPC/Topic). | S | AL-63 |
| | T365 | Acceptance of manipulated object lists or sensor metadata resulting from insufficient data authenticity verification. | T | AL-52 |
| | T366 | Distortion of fusion results caused by the acceptance of replayed perception frames due to lack of freshness checks. | T | AL-19 |
| | T367 | Masquerading as a legitimate publisher via missing or misconfigured ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T368 | Tampering of perception data payloads facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T369 | Eavesdropping on perception data/metadata via plaintext transmission in IPC/Topics. | I | AL-13 |
| | T370 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T371 | Exposure of sensitive information (object tracks, confidence scores, location estimates) due to excessive data inclusion. | I | AL-13 |
| | T372 | Fusion input delays or data drops caused by frame/message flooding saturating queues/ring buffers. | D | AL-51 |
| | T373 | Service degradation in the upper pipeline caused by a storm of error frames or unhandled exceptions. | D | AL-51 |
| DF2.2. a-priori Perception Data | T374 | Injection of a-priori/GPS/IMU frames via unauthorized processes due to weak endpoint constraints (Topic/IPC binding). | S | AL-63 |
| | T375 | Acceptance of manipulated, downgraded, or replayed Map/Pose/PVT data due to insufficient verification. | T | AL-52, AL-19 |
| | T376 | Masquerading as a legitimate publisher via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T377 | Tampering of payload facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T378 | Eavesdropping on HD Map/Path/Pose metadata via plaintext transmission in internal channels. | I | AL-13 |
| | T379 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T380 | Exposure of sensitive information (past paths, POIs, quality metrics) due to excessive data inclusion. | I | AL-13 |
| | T381 | Input delays or drops caused by Map/Message flooding saturating queues/buffers. | D | AL-51 |
| | T382 | Service degradation in the upper pipeline caused by a storm of error frames or retries. | D | AL-51 |
| DF2.3. Localization Data | T383 | Injection or tampering of localization frames via unauthorized processes due to weak endpoint constraints (Topic/IPC binding). | S | AL-63 |
| | T384 | Acceptance of manipulated or replayed Pose/Velocity/Time data resulting from insufficient verification. | T | AL-52, AL-19 |
| | T385 | Masquerading as a legitimate publisher via missing or misconfigured ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T386 | Tampering of localization payloads facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T387 | Propagation of upstream GNSS spoofing effects through the localization data flow. | T | AL-1, AL-32 |
| | T388 | Eavesdropping on current/past Pose/Velocity/Quality metrics via plaintext transmission in internal channels. | I | AL-13 |
| | T389 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T390 | Exposure of sensitive information (Precise Pose, Path IDs, Quality Metrics) due to excessive data inclusion. | I | AL-13 |
| | T391 | Fusion input delays or data drops caused by message/frame flooding saturating queues/buffers. | D | AL-51 |
| | T392 | Service degradation in the upper pipeline caused by a storm of error frames or retries. | D | AL-51 |
| | T393 | Injection or tampering of fusion results via unauthorized processes due to weak endpoint constraints (Topic/IPC). | S | AL-63 |
| | T394 | Acceptance of manipulated or replayed objects/tracks resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T395 | Masquerading as a legitimate publisher via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |

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| DF2.4. Fusion Data | T396 | Tampering of fusion payloads facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T397 | Propagation of upstream sensor spoofing/adversarial inputs into the fusion result (Ghost Objects). | T | AL-33, AL-36, AL-38 |
| | T398 | Eavesdropping on objects/tracks/confidence scores via plaintext transmission in internal channels. | I | AL-13 |
| | T399 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T400 | Exposure of sensitive information (Track IDs, Path, Environment) due to excessive data inclusion. | I | AL-13 |
| | T401 | Input delays or drops caused by message flooding or oversized payloads saturating queues. | D | AL-51 |
| | T402 | Pipeline degradation caused by a storm of error frames or retries. | D | AL-51 |
| DF3.1. Sensor Log Data | T403 | Injection or tampering of log records via unauthorized processes due to weak endpoint constraints (Socket/IPC/Topic). | T | AL-63 |
| | T404 | Acceptance of manipulated or replayed log records resulting from insufficient data authenticity/freshness verification. | T | AL-52, AL-19 |
| | T405 | Tampering of logs in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T406 | Record manipulation or arbitrary code execution via unsafe deserialization of structured log records in the collector. | T | AL-18 |
| | T407 | Eavesdropping on sensitive logs (location/tracks/IDs) via plaintext transmission in internal channels. | I | AL-13 |
| | T408 | Interception and decryption risks resulting from weak encryption or key management errors between agent and storage. | I | AL-13 |
| | T409 | Evidence loss (data drop) caused by log flooding or high-frequency transmission saturating collector queues/IO. | D | AL-51 |
| | T410 | Pipeline service degradation caused by a storm of error/exception records leading to retry accumulation. | D | AL-51 |
| DF3.2. Planning Log Data | T411 | Injection or tampering of planning log records via unauthorized processes due to weak endpoint constraints. | T | AL-63 |
| | T412 | Acceptance of manipulated or replayed log records (candidate paths/costs) resulting from insufficient verification. | T | AL-52, AL-19 |
| | T413 | Tampering of logs in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T414 | Record manipulation or arbitrary code execution via unsafe deserialization of structured log records in the collector. | T | AL-18 |
| | T415 | Eavesdropping on sensitive decision logic/paths/confidence scores via plaintext transmission. | I | AL-13 |
| | T416 | Interception and decryption risks resulting from weak encryption or key management errors between agent and storage. | I | AL-13 |
| | T417 | Evidence loss (data drop) caused by log flooding or high-frequency transmission saturating collector queues/IO. | D | AL-51 |
| | T418 | Pipeline service degradation caused by a storm of error/exception records leading to retry accumulation. | D | AL-51 |
| DF3.3. HMI Log Data | T419 | Injection or tampering of HMI log records via unauthorized processes due to weak endpoint constraints (Socket/IPC/Topic). | T | AL-63 |
| | T420 | Acceptance of manipulated or replayed log records (alert history) resulting from insufficient data authenticity/freshness verification. | T | AL-52, AL-19 |
| | T421 | Tampering of logs in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T422 | Record manipulation or arbitrary code execution via unsafe deserialization of structured log records in the collector. | T | AL-18 |
| | T423 | Eavesdropping on sensitive logs (driver voice/text/alerts) via plaintext transmission in internal channels. | I | AL-13 |
| | T424 | Interception and decryption risks resulting from weak encryption or key management errors between agent and storage. | I | AL-13 |
| | T425 | Evidence loss (data drop) caused by log flooding or high-frequency transmission saturating collector queues/IO. | D | AL-51 |
| | T426 | Pipeline service degradation caused by a storm of error/exception records leading to retry accumulation. | D | AL-51 |
| DF3.4. Mode Manager Data | T427 | Injection or tampering of mode/handover logs via unauthorized processes due to weak endpoint constraints. | T | AL-63 |
| | T428 | Acceptance of manipulated or replayed mode transition logs resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T429 | Tampering of logs in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T430 | Record manipulation or arbitrary code execution via unsafe deserialization of structured log records in the collector. | T | AL-18 |
| | T431 | Eavesdropping on sensitive logs (authority status, handover events) via plaintext transmission. | I | AL-13 |

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| | T432 | Interception and decryption risks resulting from weak encryption or key management errors between agent and storage. | I | AL-13 |
| | T433 | Evidence loss (data drop) caused by log flooding or high-frequency transmission saturating collector queues. | D | AL-51 |
| | T434 | Pipeline service degradation caused by a storm of error/exception records leading to retry accumulation. | D | AL-51 |
| DF3.5. Control Log Data | T435 | Injection or tampering of control log records via unauthorized processes due to weak endpoint constraints. | T | AL-63 |
| | T436 | Acceptance of manipulated or replayed log records (commands/ACKs/gains) resulting from insufficient verification. | T | AL-52, AL-19 |
| | T437 | Tampering of logs in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T438 | Record manipulation or arbitrary code execution via unsafe deserialization of structured log records in the collector. | T | AL-18 |
| | T439 | Eavesdropping on control gains/limits/authority status via plaintext transmission in internal channels. | I | AL-13 |
| | T440 | Interception and decryption risks resulting from weak encryption or key management errors between agent and storage. | I | AL-13 |
| | T441 | Evidence loss (data drop) caused by log flooding or high-frequency transmission saturating collector queues. | D | AL-51 |
| | T442 | Pipeline service degradation caused by a storm of error/exception records leading to retry accumulation. | D | AL-51 |
| DF3.6. MLLM Log Data | T443 | Injection or tampering of MLLM log records via unauthorized processes due to weak endpoint constraints. | T | AL-63 |
| | T444 | Acceptance of manipulated or replayed log records (system prompts/outputs) resulting from insufficient verification. | T | AL-52, AL-19 |
| | T445 | Tampering of logs in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T446 | Record manipulation or arbitrary code execution via unsafe deserialization of structured log records in the collector. | T | AL-18 |
| | T447 | Eavesdropping on sensitive logs (user voice, system prompts, command codes) via plaintext transmission. | I | AL-60, AL-13 |
| | T448 | Interception and decryption risks resulting from weak encryption or key management errors between agent and storage. | I | AL-13 |
| | T449 | Evidence loss (data drop) caused by log flooding or high-volume transmission (long prompts/large outputs). | D | AL-8, AL-51 |
| | T450 | Pipeline service degradation caused by a storm of error/exception records leading to retry accumulation. | D | AL-51 |
| DF4.1. Text/Audio Data | T451 | Injection or tampering of text/audio frames via unauthorized processes due to weak endpoint constraints (IPC/Topic). | S | AL-63 |
| | T452 | Acceptance of manipulated or replayed voice/text inputs resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T453 | Tampering of messages in transit facilitated by the absence or misconfiguration of integrity protection. | T | AL-19 |
| | T454 | Malicious modification of content via Prompt Injection, context contamination, or adversarial audio triggers. | T | AL-3, AL-38 |
| | T455 | Eavesdropping on user voice/text/metadata (session/time) via plaintext transmission. | I | AL-13 |
| | T456 | Interception and decryption risks resulting from weak encryption or key management errors between Infotainment and MLLM. | I | AL-13 |
| | T457 | Exposure of sensitive context (PII, intent, location) due to excessive data inclusion. | I | AL-60, AL-13 |
| | T458 | Delay or drop of MLLM responses caused by flooding with long prompts or high-frequency audio. | D | AL-8, AL-51 |
| DF4.2. Driver Command Code | T459 | Injection or tampering of Driver Command Codes via unauthorized processes due to weak endpoint constraints (IPC/Topic). | S | AL-63 |
| | T460 | Acceptance of manipulated or replayed command codes resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T461 | Masquerading as a legitimate publisher via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T462 | Tampering of command code payloads facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T463 | Propagation of malicious outputs from upstream Prompt Injection due to lack of schema validation. | T | AL-3, AL-43 |
| | T464 | Eavesdropping on driver intent/command codes/session metadata via plaintext transmission. | I | AL-13 |
| | T465 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T466 | Exposure of sensitive information (intent, policy flags) due to excessive data inclusion. | I | AL-13 |
| | T467 | Planning input delays or drops caused by high-frequency command flooding saturating queues. | D | AL-51 |

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| DF5. HMI Status & Alerts | T468 | Injection or tampering of alert/status messages via unauthorized processes due to weak endpoint constraints (IPC/Topic). | S | AL-63 |
| | T469 | Acceptance of manipulated or replayed HMI alerts resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T470 | Masquerading as a legitimate publisher via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T471 | Tampering of alert messages facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T472 | Session hijacking or fake session creation resulting from vulnerable session/key management. | S | AL-14 |
| | T473 | Injection of false audio warnings (e.g., "AEB Off") via synthesized or modulated voice attacks. | S | AL-54, AL-63 |
| | T474 | Eavesdropping on driving authority/vehicle status/alert content via plaintext transmission. | I | AL-13 |
| | T475 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T476 | Exposure of sensitive information (modes, alerts, session IDs) due to excessive data inclusion. | I | AL-13 |
| | T477 | HMI rendering saturation or message drops caused by alert flooding or high-frequency updates. | D | AL-51 |
| | T478 | User alert delay or suppression caused by a storm of error/exception messages. | D | AL-51 |
| DF6. Control Data | T479 | Injection or tampering of control data via unauthorized processes due to weak endpoint constraints (Device Node/IPC/Topic). | S | AL-63 |
| | T480 | Acceptance of manipulated or replayed control commands (torque/braking/path) resulting from insufficient verification. | T | AL-52, AL-19 |
| | T481 | Masquerading as a legitimate publisher/subscriber via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T482 | Tampering of control payloads in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T483 | Eavesdropping on control gains/limits/authority status via plaintext transmission in internal channels. | I | AL-13 |
| | T484 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T485 | Exposure of sensitive information (driving strategy, mode, path IDs) due to excessive data inclusion. | I | AL-13 |
| | T486 | Control command delay or drop caused by flooding, excessive frequency, or oversized payloads saturating queues. | D | AL-51 |
| | T487 | Service degradation of the control path caused by a storm of error/retry messages leading to timeout accumulation. | D | AL-51 |
| DF7. Vehicle Status | T488 | Injection or tampering of status frames via unauthorized processes due to weak endpoint constraints (Device Node/IPC/Topic). | S | AL-63 |
| | T489 | Acceptance of manipulated or replayed Gateway/Actuator status resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T490 | Masquerading as a legitimate publisher/subscriber via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T491 | Tampering of status payloads in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T492 | Eavesdropping on controller status/DTCs/authority status via plaintext transmission in internal channels. | I | AL-13 |
| | T493 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T494 | Exposure of sensitive information (VIN, ECU IDs, driving history) due to excessive data inclusion. | I | AL-13 |
| | T495 | Status delay or drop caused by frame flooding or abnormal cycles saturating OS/Driver/HMI queues. | D | AL-51 |
| | T496 | Mode decision delay caused by a storm of error/alert events leading to retry/timeout accumulation. | D | AL-51 |
| DF8. Control Signal | T497 | Injection or tampering of control frames (torque/steering) by 3rd parties due to lack of Endpoint/ID whitelisting. | T | AL-63 |
| | T498 | Acceptance of manipulated or replayed control frames resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T499 | Tampering of signals on the physical line (CAN/Ethernet) due to absence or misconfiguration of message authentication. | T | AL-19 |
| | T500 | Eavesdropping on control gains/limits/authority status via plaintext transmission (CAN/DoIP). | I | AL-13 |
| | T501 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T502 | Exposure of sensitive information (strategy, path IDs) due to excessive inclusion in frames. | I | AL-13 |
| | T503 | Bus saturation causing control delay/drop due to flooding, arbitration abuse, or oversized payloads. | D | AL-51 |

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| | T504 | Control loss or delay caused by a storm of error frames or retries accumulating timeouts. | D | AL-51 |
| DF9. Processed Sensor Data | T505 | Injection or tampering of processed results (objects/tracks) via unauthorized processes due to weak endpoint constraints. | S | AL-63 |
| | T506 | Acceptance of manipulated or replayed processed data resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T507 | Masquerading as a legitimate publisher via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T508 | Tampering of processed data payloads facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T509 | Propagation of upstream adversarial inputs or sensor spoofing (Ghost Objects) into the processed data flow. | T | AL-33, AL-36, AL-38 |
| | T510 | Eavesdropping on object/track IDs, attributes, and confidence scores via plaintext transmission. | I | AL-13 |
| | T511 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T512 | Exposure of sensitive information (path/behavior inference, identifiers) due to excessive data inclusion. | I | AL-13 |
| | T513 | Input queue/ring buffer saturation causing delays or drops due to message flooding or oversized payloads. | D | AL-51 |
| | T514 | Service degradation of Planning/Mode Manager caused by a storm of error frames or retries. | D | AL-51 |
| DF10. Planned Trajectory | T515 | Injection or tampering of trajectory paths (waypoints/speed profiles) via unauthorized processes due to weak endpoint constraints (IPC/Topic). | S | AL-63 |
| | T516 | Acceptance of manipulated or replayed planned trajectories resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T517 | Masquerading as a legitimate publisher/subscriber via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T518 | Tampering of trajectory payloads in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T519 | Eavesdropping on future paths/speed profiles/constraints via plaintext transmission in internal channels. | I | AL-13 |
| | T520 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T521 | Exposure of sensitive information (Path IDs, Precision Location, Policy Flags) due to excessive data inclusion. | I | AL-13 |
| | T522 | Input queue saturation causing delays or drops due to trajectory message flooding or oversized payloads. | D | AL-51 |
| | T523 | Service degradation of control/mode paths caused by a storm of error/retry/recalculation events leading to timeouts. | D | AL-51 |
| DF11. Manage Status | T524 | Injection or tampering of authority status messages via unauthorized processes due to weak endpoint constraints (IPC/Topic). | S | AL-63 |
| | T525 | Acceptance of replayed or downgraded Manage Status resulting from insufficient authenticity/freshness verification. | T | AL-52, AL-19 |
| | T526 | Masquerading as a legitimate publisher/subscriber via missing ROS2/DDS Authentication/Access Control. | S | AL-63 |
| | T527 | Tampering of status payloads in transit facilitated by the absence or misconfiguration of message authentication. | T | AL-19 |
| | T528 | Session hijacking or masquerading as a legitimate node due to vulnerable session/key management. | S | AL-14 |
| | T529 | Eavesdropping on driving authority/mode/alert flags via plaintext transmission in internal channels. | I | AL-13 |
| | T530 | Interception and decryption risks resulting from weak encryption or key management errors. | I | AL-13 |
| | T531 | Exposure of sensitive information (Authority Status, Handover Request) due to excessive data inclusion. | I | AL-13 |
| | T532 | Input queue saturation causing mode propagation delays due to status message flooding or oversized payloads. | D | AL-51 |
| | T533 | Service degradation of Control/Planning caused by a storm of error/retry events leading to timeouts. | D | AL-51 |